

Lesson 3: Lists Practice

45 minutes

Overview

Practice the basics of lists including creating lists and accessing, inserting, and removing elements from lists.

Standards

Full Course Alignment

CSP Conceptual Framework

- ▶ **AAP-1** - To find specific solutions to generalizable problems, programmers represent and organize data in multiple ways.
- ▶ **AAP-2** - The way statements are sequenced and combined in a program determines the computed result. Programs incorporate iteration and selection constructs to represent repetition and make decisions to handle varied input values.

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming

Agenda

Warm Up (5 minutes)

Quick Warm Up

Activity (35 minutes)

Practice Time

Wrap Up (5 minutes)

Objectives

Students will be able to:

- Accurately use list operations including accessing, inserting, and removing elements
- Correctly set up a list in a program
- Debug programs with lists

Preparation

- Review the slide on string indexes which you'll cover at the beginning of the lesson
- Review the programming progression to understand what students will be asked to do.
- Check the "**Teacher's Lounge**" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our **Virtual Lesson Modifications**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **CSP Unit 6 - Lesson 3 - Lists Practice**
 - Slides [▼ Make a Copy](#)

Teaching Guide

Warm Up (5 minutes)

Quick Warm Up

Remarks

Today we're going to have a chance to practice programming with a lot of the concepts and patterns we've explored over the last two lessons. I encourage you to go through these with a partner, but pay close attention to what each other is doing. You will be choosing one of three apps to help develop today. While the design of each of these apps is complete, it will be your job to use your knowledge of lists in order to make the apps function as intended.

Whichever app you choose, please continue to choose that **same** app on each of the choice levels. There are a total of *ten* levels for each app.

In our next lesson you're going to have to use a lot of these skills on an independent project, and these activities are good practice for what you'll find there! Alright, let's get to it!

Teaching Tip

Move Quickly to the Activity: There's a lot in the main activity of today's lesson. It is advisable to give students more time to get hands on with some code.

Activity (35 minutes)

Practice Time (35 minutes)

Group: It is recommended that students work in pairs for this lesson. Consider using pair programming, having drivers and navigators switch every 3 minutes, not every level.

 **Do This:** Direct students to Code Studio, Lesson 3 Level 1.



1-10

App Development: Lists in Action!

1

2

3

4

5

6

7

8

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Assessment Opportunity

Create Performance Task Written Response Practice

In order to help prepare students for the new component of the AP Create Performance Task, there will be several opportunities throughout the curriculum to practice creating written responses aligned with the learning objectives that might be assessed on the exam.

Sample Prompt: Identify the number of times the body of your iteration statement will execute. Describe a condition or error that would cause your iteration statement to not terminate and cause an infinite loop. If no such condition or error exists, explain how the loop could be modified to cause an infinite loop.

Wrap Up (5 minutes)

 **Discuss:** *What aspects of working with lists do you feel like clicked today? What do you still feel like you have trouble with?*

Discussion Goal: Use this opportunity to address any lingering questions or misconceptions in the room. You can also use this as a source of discussion topics to kick off the following lesson. As you lead the discussion, call out the many resources students have access to help when they're getting stuck.

Remarks

Lists can be a little bit tricky, but I saw a lot of good progress today in nailing down this concept. We may have a few lingering questions, but you also have a lot of resources available. Next time you'll have a chance to put all this together by programming an app that starts with "the blank screen"!



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